

RAINFALL at Waterfoot Cragg Holes ResCounty of South LancashireObserver W. H. Elean Esq C.E.River Basin Mersey Swell 3Waterfoot Esq Boro Esq (1917)

Rain Gauge, by _____

Diameter ...						5						
Height						10						
Altitude ...						1426		1425				
Year ...	1910	1911	1912	1913	1914	1915	1916	1917	1918	1919	MEANS.	
January ...						7.17	4.15	2.24	4.10	6.00		
February ...						7.19	5.41	1.46	7.31	1.35		
March						1.75	3.26	2.51	2.87	5.57		
April						1.88	3.22	1.56	1.05	2.83		
May						2.23	2.97	2.73	3.29	2.12		
June						.95	3.53	2.95	1.80	1.62		
July						5.50	2.95	2.72	4.98	2.45		
August						5.08	5.36	8.36	3.69	4.88		
September ..						.84	2.79	3.11	13.50	2.87		
October ...						2.95	9.98	9.83	4.87	3.13		
November ..						2.39	4.24	4.51	3.09	3.79		
December...						9.72	3.56	2.39	10.06	8.71		
TOTALS ...						47.65	51.42	44.37	60.61	45.32		

1917-19 used 20.1.22. Feb. 4.22.

Rejected from Survey. 4" low.

Date.	Lat. N.	Long. W.	Railway Station.	Distance.	Direction.	Church.	Distance.	Direction.
	53° 46.20'	2° 44.84'	Waterfoot	1 3/8 m	S	Newchurch	2 1/8	N S.

2910
20
(2910/1)RAINFALL at Waterfoot - Gragg Hole ReservoirCounty of LancashireObserver H. H. Newton Esq. C.E.River Basin Mersey - Irwell &F. W. Mozley, Esq. C.E. (1925)L.O. Pattern Rain Gauge, by
(126)Borough & Water Eng. (Bacup)

Diameter ...		5									
Height		1	0								
Altitude ...		1425				1323					1325'
Year	1920	1921	1922	1923	1924	1925	1926	1927	1928	1929	1930. MEANS.
January ...	4.85	7.67	4.28	3.73	4.14	4.27	6.29	4.60	11.68	3.25	6.81
February ...	3.49	.53	5.73	6.79	1.28	7.98	4.63	2.81	5.04	1.54	.95
March	4.36	4.22	4.29	2.63	2.50	1.76	2.55	4.99	5.23	1.49	3.18
April	6.71	1.64	3.38	5.00	2.96	3.37	1.90	3.37	1.60	1.36	3.91
May	6.97	4.30	2.79	4.28	7.58	7.21	4.47	2.41	1.43	3.83	1.77
June	3.16	.40	2.93	1.23	3.17	1.14	3.17	4.99	8.55	1.77	1.81
July	9.71	2.25	6.64	4.87	6.17	2.17	3.32	5.30	2.50	3.46	6.25
August	3.71	8.23	6.84	7.31	10.12	5.55	5.77	9.50	7.16	5.34	8.15
September...	2.88	2.19	5.55	6.17	6.74	4.73	4.60	6.51	0.78	2.05	6.13
October	2.06	3.95	1.10	7.89	7.19	7.31	5.08	5.49	8.91	7.14	6.40
November...	3.09	3.96	3.69	7.83	3.39	2.84	9.20	5.16	8.11	10.10	7.52
December...	5.08	6.48	5.49	5.29	6.39	3.88	1.53	2.45	3.91	8.08	4.76
TOTALS...	56.07	45.82	52.71	63.02	61.63	51.21	52.50	58.88	65.20	49.10	57.14

1917-20 recd. 20.1.22. also x1.22

Rejected from survey 4" low.

Position of Station on reduced Ordnance Survey
Map, 2 miles to 1 inch Series

No. of Map

Reference No.

Date.	Lat. N.	Long. W.	Railway Station.	Distance from gauge	Direction	Parish Church.	Distance from gauge	Direction
1921	53° 40' 20"	2° 14' 0"	Waterfoot	1 3/8 mi	S	Newchurch	2 1/8 mi	S.
1927			Bacup	3/4 m.	S.	Bacup	1 m.	S. E.

RAINFALL at Waterfoot (Cragg Reservoir)County of South LancashireObserver E. W. Motley Esq. C.E.River Basin Mersey-Trent-3Bacup Borough Council (1936)Type of Gauge No. 2Annual Average 47.3 in. based on 18 years obs. un-weighted. Period of average 1881-1915 Ref P.B. 355

706862/26.

Diameter	5.										
Height	1.0										
Altitude	1325.										
Year	1931	1932	1933	1934	1935	1936	1937	1938	1939	1940	Means
January	5.18	5.85	3.37		1.91	6.22	4.02				
February	6.35	3.2	3.52		6.17	3.16	5.93				
March	7.4	3.05	3.73		1.44	3.35	3.11				
April	3.86	4.21	2.39		4.40	1.60	3.57				
May	8.77	5.88	2.66		1.46	8.9	2.29				
June	7.17	1.62	2.22	3.19	4.87	5.17	2.44				
July	6.64	5.23	4.17	2.38	2.09	5.42	2.33				
August	7.54	1.53	1.97	3.99	2.24	3.77	3.46				
September	3.66	5.62	2.11	4.44	9.58	5.55	2.33				
October	2.79	2.33	4.79	5.53	7.30	3.92	2.45				
November	11.77	3.62	2.83	2.55	6.00	5.68	2.12				
December	2.24	3.55	7.4	9.23	4.34	5.21	5.10				
TOTALS	60.61	48.81	34.50	—	50.80	49.94	40.20				
	2	3	3		20/10	20/10	20/10				

1936. Readings taken for comparison with new g. No 2910/1

1935. r/p. Check

1936. " "

1937. r/p. check Record T. 12/37

Position of Station on reduced Ordnance Survey Map.
1 mile to 1 inch Series

No. of Map.....

Reference No.....

Date	Lat. N.	Long.	Railway Station.	Distance from gauge	Direction from gauge	Parish Church.	Distance from gauge	Direction from gauge
1931			Waterfoot	1 3/8 m	N.W.	Newchurch	2 1/2 m	N.W.

RAINFALL at Waterfoot (Crags Reservoir) reservoir.County of N. Lancs. Observer Bacup Borough Council.River Basin 3-Murray-Sussex.Type of Gauge M.O.

Annual Average.....in. based on.....years obs. un-weighted. Period of average.....Ref.....

C 1934

706863/26.

Diameter				5"							
Height				1' 0"							
Altitude				1325'							
Year	1931	1932	1933	1934	1935	1936	1937	1938	1939	1940	Means
January				4.72	2.27	6.85	5.00	7.20	8.95	3.05	
February				.62	7.31	3.43	7.42	2.65	6.88	3.20	
March				3.93	1.85	4.03	3.59	1.75	3.46	3.89	
April				3.55	4.93	1.76	4.23	.80	2.41	4.29	
May				4.37	1.63	.92	2.58	4.90	2.21	1.25	
June				3.62	5.27	5.37	2.81	5.80	4.57	.93	
July				2.77	2.44	6.02	3.26	5.68	9.17	4.64	
August				4.57	2.51	4.32	3.84	5.22	3.68	2.66	
September				5.05	10.60	6.11	2.63	3.20	1.81	5.82	
October				6.74	9.89	5.60	3.13	13.22	3.19	6.49	
November				3.03	6.56	6.64	2.33	8.54	8.75	10.34	
December				9.76	5.32	6.88	5.41	7.21	4.76	5.98	
TOTALS				52.73	60.58	67.93	46.23	66.17	59.83	52.54	
				D	D	D	D	D	D	D	

This site to replace 2910 which was rejected from survey as too low.

Position of Station on reduced Ordnance Survey Map, } No. of Map..... Reference No.....
1 mile to 1 inch Series

Date.	Lat. N.	Long.	Railway Station.	Distance from gauge	Direction from gauge	Parish Church.	Distance from gauge	Direction from gauge
1934			Waterfoot	1 3/8 m	N.W.	Cheshchurch	2 1/8 m	N.W.

Station No. 2910/1

RAINFALL at Waterfoot (Gragg Reservoir)

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County of S. Lancs

Observer Bacup Borough Council.

River Basin Mersey - Irwell - 3

Type of Gauge M.O. Snowden 1950

Annual Average in. based on years obs. un-weighted. Period of average Ref.

C. 1934

File No. 706362/36

Diameter	5"			560	705						
Height	1'0"				1						
Altitude	1325'				41						
Year	1941	1942	1943	1944	1945	1946	1947	1948	1949	1950	Means
January	2.76	3.47	8.29	7.13	4.59	6.80	5.48	13.04	3.76	4.43	
February	7.02	3.00	3.94	2.53	6.90	6.65	3.72	4.41	4.55	7.31	
March	5.07	5.01	1.68	.87	1.82	1.77	9.20	2.26	2.98	3.58	
April	3.05	2.80	3.22	4.46	3.01	1.92	4.80	2.43	4.99	4.44	
May	3.54	4.71	4.37	3.64	5.26	1.55	2.59	2.75	3.75	2.54	
June	.69	1.13	4.76	3.56	6.31	4.86	3.44	6.36	.68	3.05	
July	5.48	6.28	3.93	4.66	3.92	4.49	4.22	3.45	2.33	7.18	
August	8.25	6.78	8.81	3.96	4.43	9.22	.26	6.21	4.89	8.35	
September	1.15	3.48	6.47	7.48	3.64	9.62	4.05	4.24	2.34	10.30	
October	6.03	7.45	4.25	6.93	7.50	1.75	1.22	3.58	6.83	3.52	
November	4.26	1.45	3.99	3.99	.67	10.25	7.75	2.83	8.39	7.51	
December	3.35	5.51	3.11	6.61	4.62	5.43	4.69	5.63	9.13	3.09	
TOTALS	50.65 D	57.07 D	56.82 D	61.82 D	52.67 D	64.31 D	51.42 D	57.19 D	54.62 D.	65.30 D	

Position of Station on reduced Ordnance Survey Map,
1 mile to 1 inch Series

No. of Map.....

Reference No.....

Date.	Lat. N.	Long.	Railway Station.	Distance from gauge	Direction	Parish Church.	Distance from gauge.	Direction
1941			Waterfoot	1 7/8 m	NW	Newchurch	2 1/8 m	NW.

BRITISH RAINFALL ORGANIZATION

Station No. 2910/1RAINFALL at Waterfoot (Cragg Reservoir)

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County of S. LancashireObserver Bacup Borough CouncilRiver Basin Mersey-Deverell - 3Type of Gauge LondonAnnual Average 53.3 in. based on 17 years obs. un-weighted. Period of average 1881-1915 Ref. 1. W.S." " 57.44 " " 20 " " within 1934-57. " " 1916-50 " 2File No. 786363/24

Diameter ...	5"			560	705						
Height ...	1' 0"				1						
Altitude ...	1325'				51						
Year ...	1951	1952	1953	1954	1955	1956	1957	1958	1959	1960	Means
January ...	6.39	6.34	-100	5.11	4.17		7.11	6.25	4.43	8.71	
February ...	4.78	1.96	-100	(2.82)	4.11		5.27	10.17	.94	7.57	
March ...	6.79	4.07	-100	3.12	2.67		6.19	3.46	3.59	1.94	
April ...	3.39	5.23	-100	.93	3.38		.80	2.13	4.33	3.59	
May ...	2.35	3.19	-100	3.75	4.90		2.61	6.89	2.23	2.71	
June ...	1.49	5.29	-100	4.54	4.91		2.58	6.01	5.11	1.63	
July ...	4.04	2.38	7.14	6.86	1.62		7.17	5.17	4.71	9.34	
August ...	6.55	6.69	6.04	8.89	1.43		7.84	4.30	.93	6.66	
September ...	4.67	5.02	5.32	6.19	4.19		7.95	7.38	.52	4.36	
October ...	1.11	6.23	3.10	11.80	2.90		8.76	6.15	6.56	7.09	
November ...	11.14	3.17	6.78	8.82	8.47		2.54	2.70	6.59	9.91	
December ...	10.95	5.30	2.59	8.87	7.01		5.89	5.63	11.35	6.01	
TOTALS ...	63.65	55.37	-100	71.70	44.76		64.71	66.24	51.29	69.52	
	D	n.p.		n.p.	d		D	D	D	D	

1952. Values copy of those for 2911/1. Readings for this station not available
 1953. Note in file that records for this station commence 1908. New gauge ordered, in operation end of Sept. n.p.m.
 1954. n.p. m. received too late. (21/11/54).
 1955. Jan. Feb. Mar. + Dec. corr'd for snow by M.O.B.
 20000

Position of Station on Ordnance One-Inch map :

No. of Map.....

National Grid Reference SD/846197

Date.	Lat. N.	Long.	Railway Station.	Distance Direction from gauge	Parish Church.	Distance Direction from gauge.
1951			Waterfoot	1 1/8 m. N.W.	Newchurch	2 1/2 m. N.W.

Summary of Earlier Records from 10-year Sheets :

Record D, or d	1910-19	1920-30	1931-40	1941-50	1951-60
			C1939 ✓	✓	✓